

The Perils of Bilateral Sovereign Debt

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Official Sovereign Debt

- A large share of sovereign borrowing takes the form of **official** debt
... Multilaterals, development banks, other governments
- Emergence of new bilateral creditors **outside** the Paris Club ▶ IDS data
... with claims to **seniority** and sometimes **confidential** terms

Questions

- How does the presence of a large senior lender affect sovereign debt markets?
- What are its welfare implications for borrowing governments?

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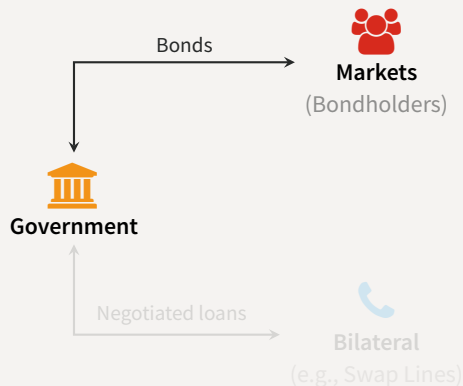
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Evaluating Senior Official Creditors

Quantitative sovereign debt model with

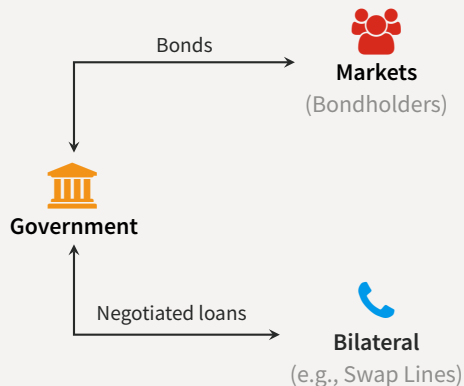
- Competitive creditors in private **markets**
- Large **bilateral** lender
 1. Superior enforcement
[de-facto seniority]
 2. Bargained terms
[price and quantity]
 3. Short-maturity loans
- Prime example: Central Bank **swap** lines
(Horn et al., 2021)
- Focus on the **interaction** between both funding sources



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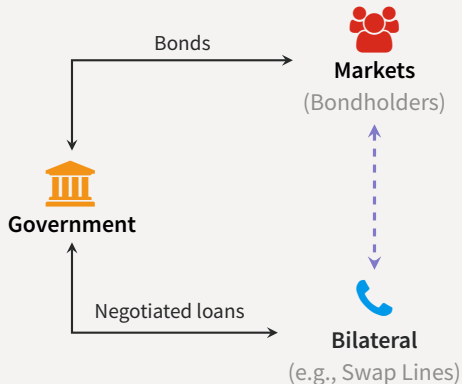
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Relational Overborrowing

Main findings

- Bilateral loans have significant effects on equilibrium outcomes
 - ... provide funding when other sources dry up (e.g. because of default risk) ▲
 - ... can also incentivize more **risk-taking** ▼
- If the rate on bilateral loans is decreasing in *market* debt [cross-elasticity]
 - ... government issues market debt more quickly, delevers more slowly
 - ... spends longer in the risky region
 - ... defaults more frequently
 - ... **welfare losses for the government**
- Cross-elasticity can emerge endogenously from **bargaining** ☞
 - ... at plausible values for bargaining weights
 - ... increased frequency of defaults dominates extra liquidity
 - ... **relational overborrowing**

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- Sovereign debt/default with interactions from ‘official’ debt
 - ... senior debt (Hatchondo, Martinez & Önder 2017), senior debt with conditionality (Boz 2011, Fink & Scholl 2016), bailout agencies (Corsetti, Guimarães & Roubini 2006, Kirsch & Rühmkorf 2017, Roch & Uhlig 2018), official debt (Dellas & Niepelt 2016, Arellano & Barreto 2024, Liu, Liu & Yue 2025)
- Data on new official creditors
 - ... Horn, Reinhart & Trebesch 2021a, 2021b, Gelpern et al. 2021, Horn, Parks, Reinhart & Trebesch 2023
- Central Bank swap lines
 - ... among advanced economies (Bahaj & Reis 2021, Cesa-Bianchi, Eguren-Martin & Ferrero 2022), data for emerging-market borrowers (Perks, Rao, Shin & Tokuoka 2021)

Roadmap

- Theory
 - Bargaining and the Cross-Elasticity
 - The Cross-Elasticity and Debt Dilution
- Quantitative Model
 - Endogenous Bargaining
 - Programming the Large Lender
- Concluding remarks

Theory

Theory

Bargaining and the Cross-Elasticity

Bargaining Induces a Cross-Elasticity

- Two periods $t \in \{0, 1\}$, endowments $y_0 < y_1$
- Government discount factor β , lender's discount factor $\beta_L > \beta$
- **Bargaining** transfer x at $t = 0$ for repayment $x(1 + r)$ at $t = 1$, weight θ for lender

$$\max_{x,r} \mathcal{L}(x,r)^\theta \times \mathcal{B}(x,r;y_0,y_1)^{1-\theta}$$

where

$$\mathcal{L}(x,r) = \underbrace{-x + \beta_L x(1+r)}_{\text{agreement}}$$
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Bargaining Induces a Cross-Elasticity

- First-order conditions for bargaining problem

$$u'(y_0 + x) = \frac{\theta}{1-\theta} \frac{\mathcal{B}(x, r; y_0, y_1)}{\mathcal{L}(x, r)}$$

Surplus sharing

$$u'(y_0 + x) = \frac{\beta}{\beta_L} u'(y_1 - x(1 + r))$$

Euler

- Can prove that

$$\frac{\partial x}{\partial y_0} \leq 0 \quad \text{and} \quad \frac{\partial r}{\partial y_0} \leq 0$$

- Later** y_0, y_1 are given by market debt dynamics, so
 - Recent issuances: increases y_0 and decreases y_1 , makes you **stronger**
 - Legacy debt: lowers y_0 and y_1 , typically makes you **weaker**

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Theory

The Cross-Elasticity and Debt Dilution

Debt Dilution with Undefaultable Loans

Issue long bonds b at $t = 0$, short bonds d and loans m at $t = 1$, repay at $t = 2$

$$c_0 = q(b + d)b \quad c_1 = q(b + d)d + \phi(b, d, m)m \quad c_2(z) = y(z) - m - \min\{h(z), b + d\}$$

with

$$q(b + d) = \mathbb{P}(b + d \leq h(z)) \quad \text{and} \quad m \leq \bar{m}$$

to maximize $\sum_{t=0}^2 \mathbb{E}[u(c_t)]$

Commitment: Choose d internalizing effect on initial prices

$$u'(c_0) \underbrace{q'(b + d)b}_{\text{past prices}} + u'(c_1) \left(\underbrace{q(b + d)}_{\text{revenue}} + \underbrace{q'(b + d)d}_{\text{current prices}} + \underbrace{\phi'_d(b, d, m)m}_{\text{cross-elasticity}} \right) = q(b + d) \mathbb{E}[u'(c_2^R)]$$

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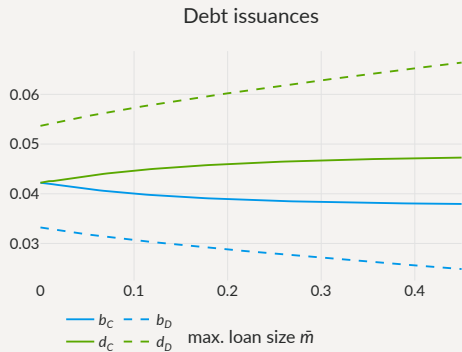
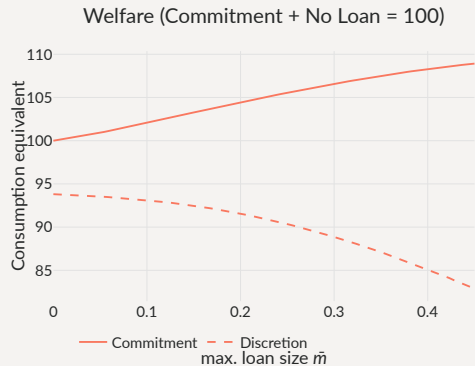
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not under discretion

Debt Dilution with Undefaultable Loans and Cross-Elasticity



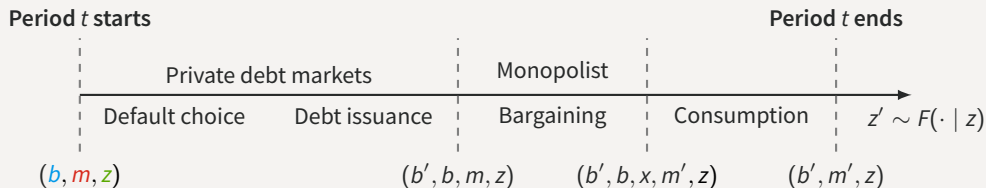
Takeaways

- Bargaining can induce a cross-elasticity from market debt to loan terms
 - ... through threat-point manipulation
- The cross-elasticity worsens the debt dilution problem
 - ... it can also create welfare losses if strong enough

Quantitative Model

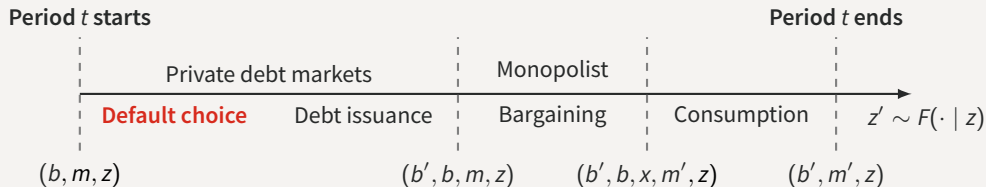
Timeline of Events

- Enter period t owing b to bondholders, m to monopolist, income $y(z)$



Timeline of Events

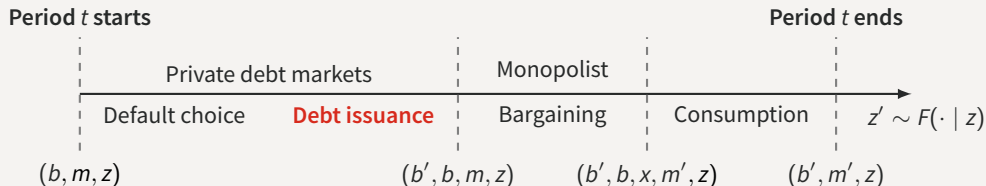
- Choose to **repay** or **default** the *market* debt subject to convex output costs



$$v(b, m, z) = \max \{ v_R(b, m, z) + \epsilon_R, v_D(m, z) + \epsilon_D \}$$

Timeline of Events

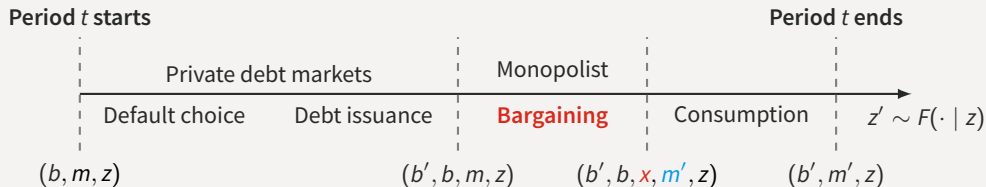
- If repaid, issue new (long-term) debt b' in markets at price q [coupon rate κ , maturity $1/\delta$]



$$q(b', b, m, z) = \beta_L \mathbb{E} [(1 - 1_{\mathcal{D}}(b', m', z')) (\kappa + (1 - \delta)q(b'', b', m', z')) | z]$$

Timeline of Events

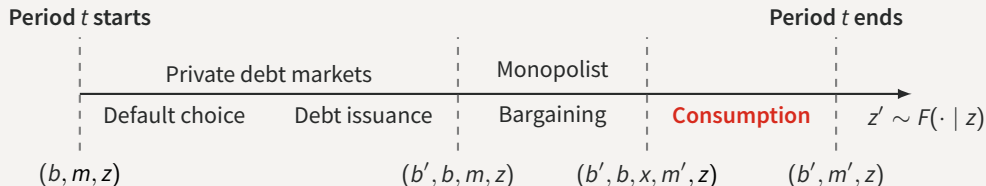
- Meet with senior lender, decide any transfers x and new/remaining balance m'



$$r(x, m', m) = \frac{m'}{x + m} - 1$$

Timeline of Events

- Consume **output** plus **revenues from debt issuance** plus **transfers** minus **debt service**



$$c_R = y(z) + q(b', b, m, z)(b' - (1 - \delta)b) + x_R(b', b, m, z) - \kappa b$$

$$c_D = y(z) - h(z) + x_D(m, z); \quad x_D(m, z) \leq \Gamma(m)$$

Calibration

- Standard strategy in sovereign debt: match Argentina 1993-2001 to model w/o swaps

	Parameter	Value
Sov's discount factor	β	0.9607
Sov's risk aversion	γ	2
Pref. scale	χ	0.025
Lender's barg. power	θ	0.5
Risk-free rate	r	0.01
Duration of debt	ρ	0.05
Income autocorr.	ρ_z	0.9484
Std. dev. y_t	σ_z	0.02
Reentry prob.	ψ	0.0385
Default cost: linear	d_0	-0.2514
Default cost: quadratic	d_1	0.292

	Data	Only market
Avg spread (bps)	815	740
Std spread (bps)	443	485
Debt-to-GDP (%)	17.4	18.1

Note moments from pre-default samples

Quantitative Model

Endogenous Bargaining

Bargaining Stage with Monopolist

- At state z , owing debt b bonds and m on the loan and having issued b'

$$\max_{x, m'} \mathcal{L}_R(b', x, m, m', z)^\theta \times \mathcal{B}_R(b', b, x, m, m', z)^{1-\theta}$$

Government surplus
Lender surplus

- Lender's surplus

$$\mathcal{L}_R(b', x, m, m', z) = \underbrace{(-x + \beta_L \mathbb{E}[h(b', m', z') | z])}_{\text{agreement}} - \underbrace{(m + \beta_L \mathbb{E}[h(b', 0, z') | z])}_{\text{threat point}}$$

- Government's surplus

$$\begin{aligned} \mathcal{B}_R(b', b, x, m, m', z) = & \underbrace{u(y(z) + B(b', b, m, z) + x) + \beta \mathbb{E}[v(b', m', z') | z]}_{\text{agreement}} \\ & - \underbrace{(u(y(z) + B(b', b, m, z) - m) + \beta \mathbb{E}[v(b', 0, z') | z])}_{\text{threat point}} \end{aligned}$$

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same sdf as markets

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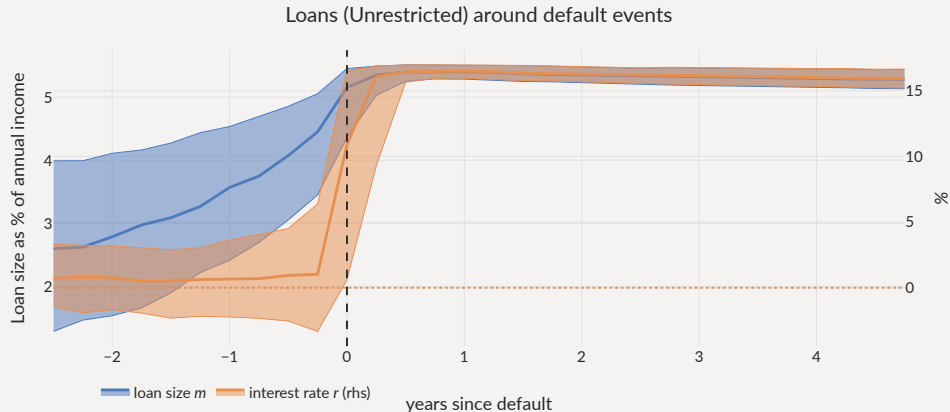
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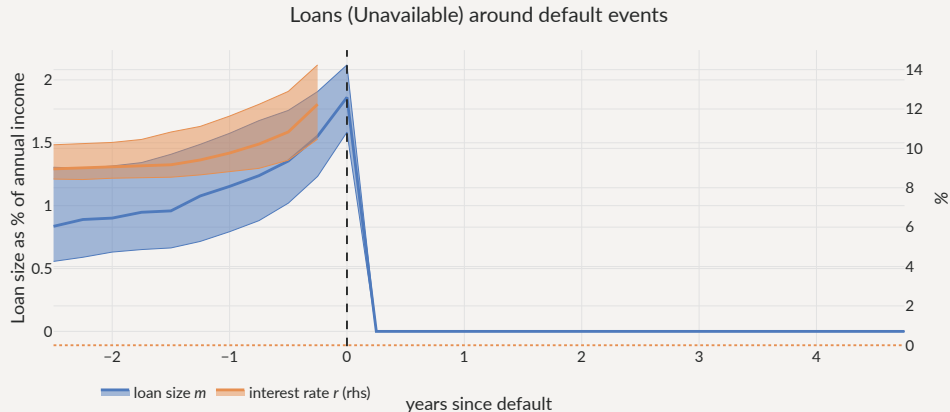
Usage of Loans

- Sharp acceleration before defaults, maxed out *in* default if possible



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Limiting Loans in Default

• **Limited:** cannot borrow in default $\Gamma(m) = m$

Unavailable: $\Gamma(m) = 0$

	Only market	Unrestricted, $\theta = 0.5$	Limited, $\theta = 0.5$	Unavailable, $\theta = 0.5$
Avg spread (bps)	630	1,398	1,209	650
Std spread (bps)	474	894	860	381
$\sigma(c)/\sigma(y)$ (%)	104	99	100	104
Debt to GDP (%)	16.4	17.6	17.6	18
Loan to GDP (%)	0	2.44	2.22	0.903
Loan spread (bps)	–	-298	-126	523
Corr. loan & spreads (%)	–	68.9	67.7	63.8
Default frequency (%)	5.03	9.55	9.04	5.28
Welfare gains (rep)	–	-0.095%	-0.038%	0.12%

Note Statistics from each ergodic distribution

Limiting Loans in Default

- Default rates:  with loans (less with Limited/Unavailable)

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- Welfare: $\approx$ with loans (Unavailable $\succcurlyeq$ Limited $\succcurlyeq$ Unrestricted)

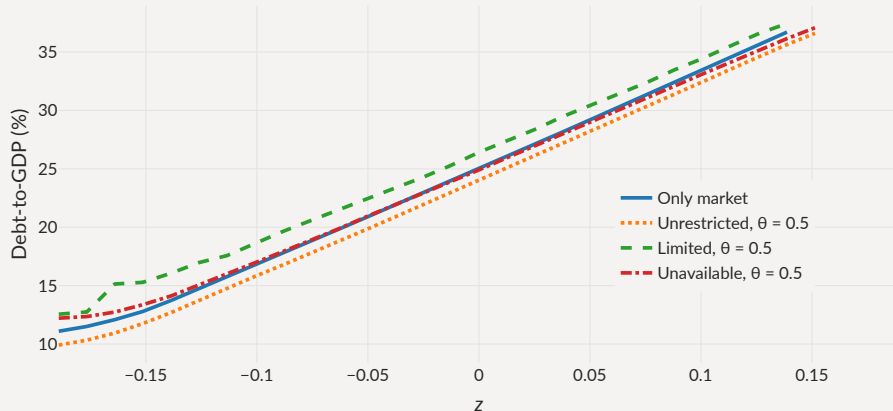
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Default Barriers with Loans

- **Unrestricted**: default barrier marginally inward, others marginally *outward*

Debt levels at which $P(b,m,z)$ crosses 50%

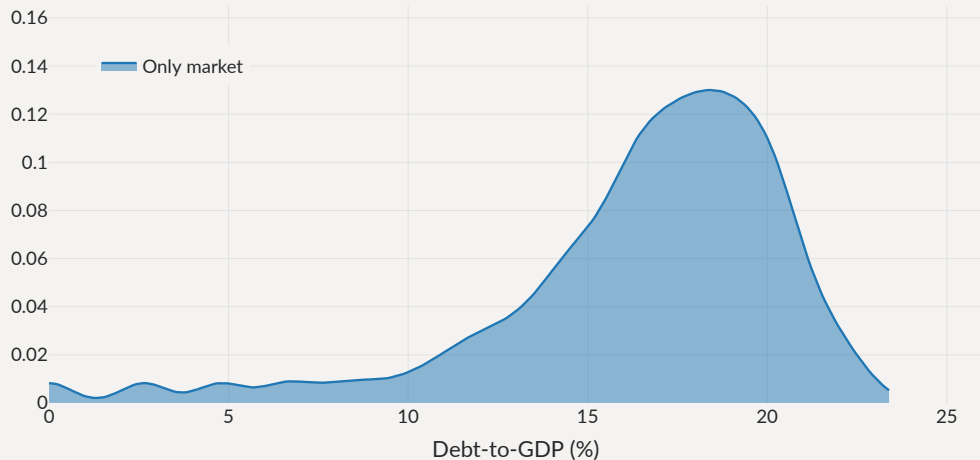


If Loans help repay the debt,

Why are there **more** defaults with loans?

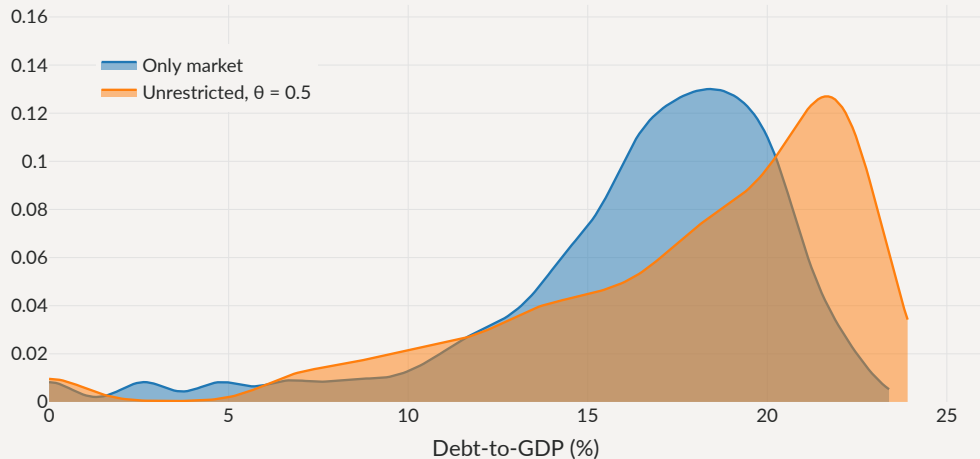
Debt Levels with Loans

Distribution of debt levels



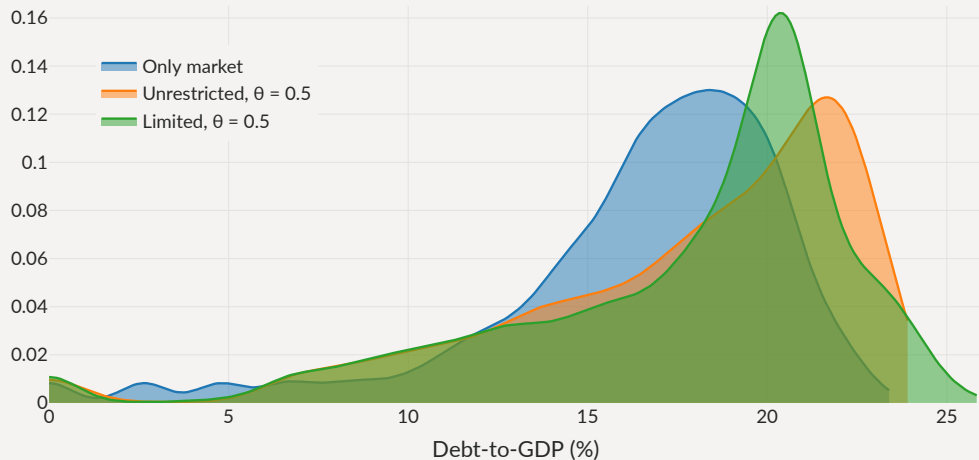
Debt Levels with Loans

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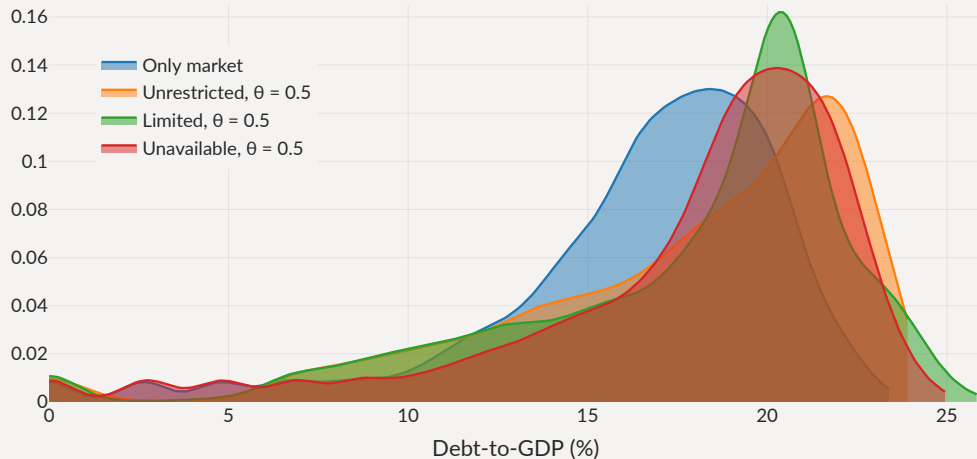
Debt Levels with Loans

Distribution of debt levels



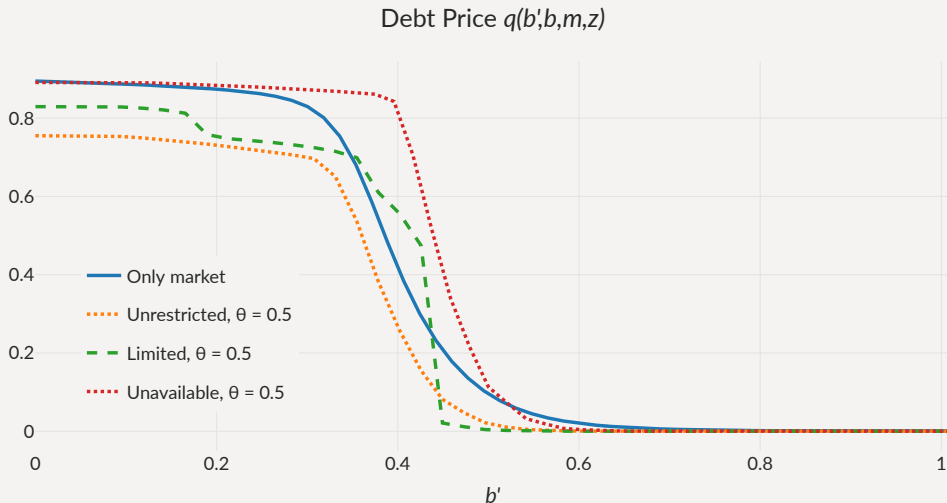
Debt Levels with Loans

Distribution of debt levels



Debt Prices with Loans

Lower prices with same ex-post default rates: **relational overborrowing** similar to debt dilution

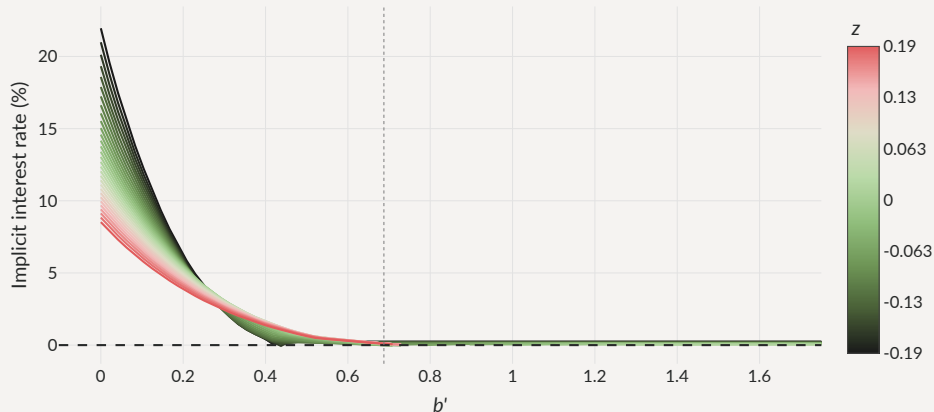


Government's surplus

$$\begin{aligned}\mathcal{B}_R(b', b, x, m, m', z) = & u(y(z) + B(b', b, m, z) + x) + \beta \mathbb{E} [v(b', m', z') \mid z] \\ & - (u(y(z) + B(b', b, m, z) - m) + \beta \mathbb{E} [v(b', 0, z') \mid z])\end{aligned}$$

- Revenues from debt issuance $B(b', b, m, z)$ modulate the value of the threat point
 - After large revenues (high q , high b'), gov't flush with cash, **strong** in bargaining
 - After bad issuance (low q or low b'), gov't **weak** in bargaining

Threat point manipulation: increase market b' to reduce bilateral r **Cross-elasticity**
Loan interest rate (Unavailable)



Government's Euler equation for bonds

$$u'(c) \left(q + \frac{\partial q}{\partial b'} i + \frac{1}{1+r} \frac{\partial m'}{\partial b'} + \frac{\partial \frac{1}{1+r_b}}{\partial b'} m' \right) = \beta \mathbb{E} \left[v_b(b', m', z') + v_m(b', m', z') \frac{\partial m'}{\partial b'} \mid z \right]$$

- An extra bond:

... yields marginal revenue $q + \frac{\partial q}{\partial b'} i$ [standard debt-Laffer curve effect]

... makes you **stronger** in bargaining now

... affects the amount m' borrowed from monopolist

... affects terms of borrowing with monopolist

... makes you **weaker** in bargaining later

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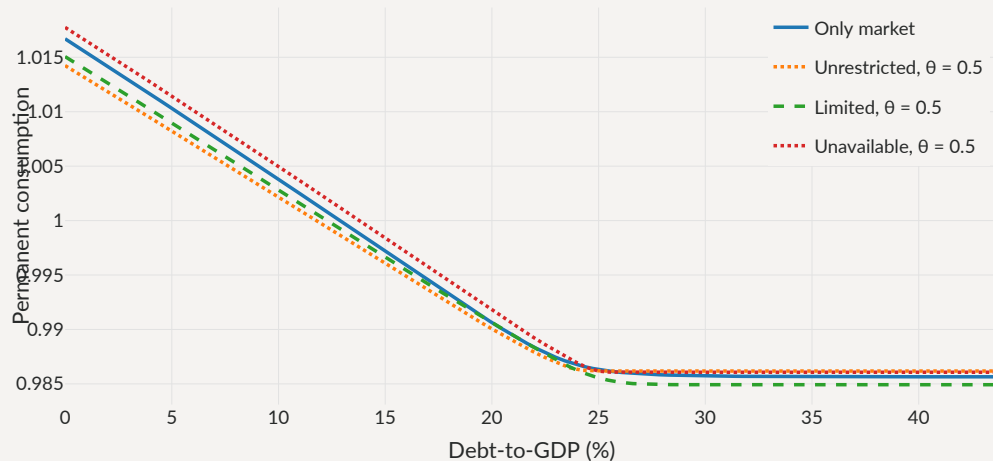
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Welfare Effects of Bilateral Loans

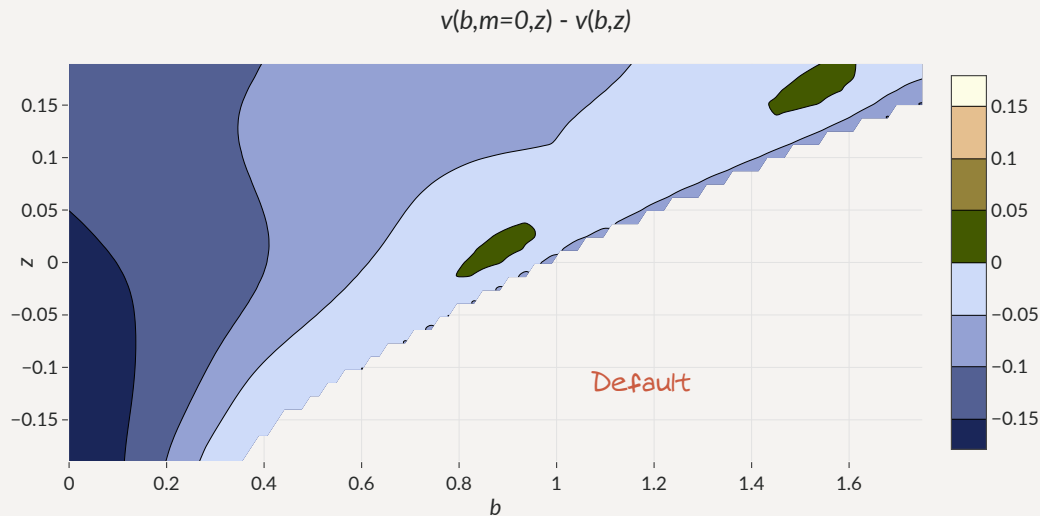
Limited $\succcurlyeq$ Unrestricted, but...

Value at $m=0$, average output



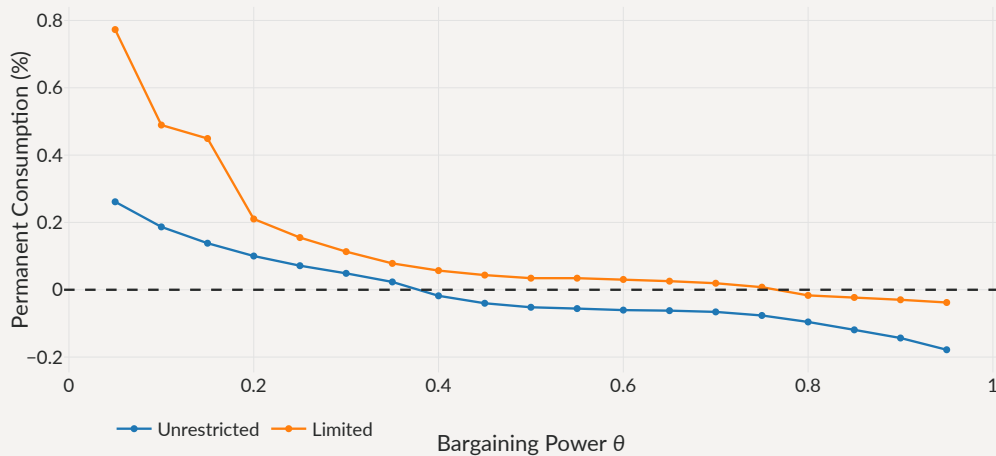
Welfare Effects of Bilateral Loans (cont'd)

Overall welfare effect small: more risk, but more frontloading in short run.



Bargaining Power and Welfare

Welfare Gains



Quantitative Model

Programming the Large Lender

Possible Rules

- Explore interest rate rules of the form

$$r(b', m') = \max\{r^*, \alpha_0 + \alpha_b b' + \alpha_m m'\}$$

- Two versions

Size-dependent

$$\alpha_0 > 0, \alpha_b = 0, \alpha_m > 0$$

Risk-inducing

$$\alpha_0 > 0, \alpha_b < 0, \alpha_m \geq 0$$

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Equilibrium with Exogenous Rules

- ‘Only market’ standard calibration to Argentina 1993-2001

	Only market	Size dependent r	Risk inducing r	Limited, $\theta = 0.5$
Avg spread (bps)	630	402	550	650
Std spread (bps)	474	247	340	381
$\sigma(c)/\sigma(y)$ (%)	104	105	104	104
Debt to GDP (%)	16.4	17.5	17	18
Loan to GDP (%)	0	0.36	0.12	0.903
Loan spread (bps)	–	704	6,535	523
Corr. loan & spreads (%)	–	51.4	1.73	63.8
Default frequency (%)	5.03	3.46	4.55	5.28
Welfare gains (rep)	–	0.25%	-0.018%	0.12%

Note Statistics from each ergodic distribution

Equilibrium with Exogenous Rules

- Default rates:  with size dependent  with risk-inducing

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Note Statistics from each ergodic distribution

Equilibrium with Exogenous Rules

- Welfare: ↑ with size dependent ↓ with risk-inducing

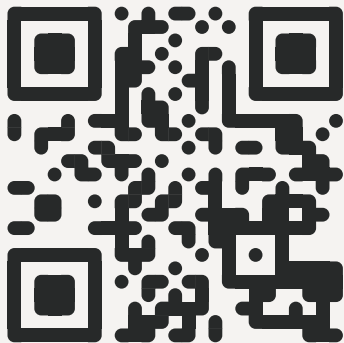
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Concluding remarks

The Perils of Bilateral Sovereign Debt

- Simple model of borrowing from **markets** and a **senior bilateral lender**
 - ... strong interaction between two markets for sovereign debt
 - ... even if bilateral loans are **not** used intensely on the equilibrium path
- **Dangerous** when bilateral interest rate responds negatively to *market* debt
 - ... cross-elasticity induces risk-taking, more defaults, welfare losses
 - ... Bargaining as an example of situation where cross-elasticity emerges
- Cross-elasticity constitutes a simple test to assess welfare gains of **new** instruments
 - ... or a boost to the gains of fiscal rules, state-contingent debt...

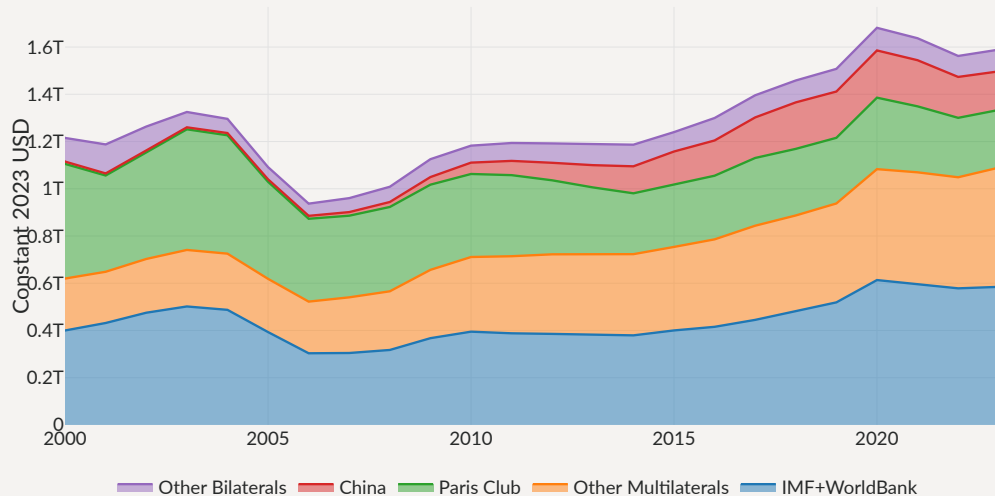


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A New Landscape for Official Sovereign Debt

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Total Official Debt



Loan drawings m' (Unavailable)

